



Lab Task:	12
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Course Code:	CL2005
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Course Title:	Database Systems - Lab
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Semester:	Spring 2026
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Instructor:	Muhammad Mehdi
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Email:	muhammad.mehdi@nu.edu.pk
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Guidelines

- Please review the lab manual before starting the task to ensure you fully understand the requirements.
- You may use **only the concepts covered in the lab manual** to complete this task.
- Your submission must be a **single PDF file** that includes:
 - The source code (in plain text)
 - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
Example: *24p-1234_ozair_labtask12.pdf*
- Unethical use of AI tools will result in a **deduction of 5 marks**.
- Late submissions will incur a penalty of **-1 mark per day**.

Tasks

You are provided with a SQL script file named *lab-12-defense_inventory.sql*, which contains SQL statements to create and populate a simplified defense inventory system database.

Execute/Import the given script file in MySQL server to load the dataset. After successfully setting up the database, perform the following tasks:

1. Create an **AFTER UPDATE** trigger on **base_inventory** that automatically inserts a record into **procurement_orders** when the updated quantity of a weapon falls below its minimum required stock, ordering the required quantity to restore the stock level.
2. Create an **AFTER INSERT** trigger on transfers that updates **base_inventory** by deducting the transferred quantity from the source base and adding it to the destination base.
3. Create a stored procedure that takes **weapon_id** and **quantity** as inputs and redistributes stock between the bases 'Alpha', 'Bravo' by transferring the **quantity** from the base with surplus stock to the base with shortage using conditional logic.



4. Create a stored procedure that takes `weapon_id` and `total_quantity` as input and distributes the procurement across bases by repeatedly adding 5 to the base with the lowest stock using a `WHILE` loop.
5. Create a stored procedure that takes a `weapon_id` as input and returns the total stock across all bases along with a status label ('`CRITICAL`' or '`STABLE`') based on whether any base has stock below the minimum required level.
6. Create a user-defined function that takes a `base_id` as input and returns '`HIGH RISK`' , '`MEDIUM`' , or '`LOW`' based on how many weapons in that base are below their minimum required stock.
7. Create a user-defined function that takes a `base_id` as input and returns the total monetary value of all weapons in that base by calculating the sum of quantity multiplied by unit cost.